A Spectral Reduction of the Collatz Conjecture via Phantom Orbit Shadowing

Piero Borgatta*
Independent Researcher

May 2026

Zenodo: 10.5281/zenodo.20021538 Source: github.com/PieroBorgatta/Collatz

Abstract

We propose a structural reformulation of the Collatz conjecture in terms of a finite-dimensional spectral problem. Starting from the empirical observation that "rebel" orbits of the Syracuse map shadow, for finite stretches, periodic rational points of the 2-adic dynamics associated to negative rational fixed points, we construct an episode graph on the space of phantom orbits. The strongly connected component (SCC) of this graph that supports the slowest descent is encoded as a finite weighted transfer operator on a refined phase state $(v_2(t), \mathrm{odd}(t) \bmod 4, h \bmod 4)$ at the critical node ($k = 12$, $c = 2$, $b = 1$).

We prove an exact congruential shadowing lemma reducing the question of arbitrarily long shadowing episodes to a residue condition $n \equiv q_w \pmod{2^{bA+1}}$, where q_w is the 2-adic representative of a phantom word w and $A = |w|_{\nu_2}$ its period sum. We then build, for each truncation depth T and high-bit lift count j, a finite matrix $\mathrm{FULL}_{T,j}$ encoding first-return behavior at the critical node, decompose it as $\mathrm{FULL}_{T,j} = \mathrm{CORE}_{T,j} + \mathrm{TAIL}_{T,j}$, and apply a weighted Collatz–Wielandt bound with respect to the right Perron eigenvector v of $\mathrm{FULL}_{T,j}$. Our numerical results show

$$\rho(\mathrm{FULL}_{T,j}) \leq \max_i(\mathrm{CORE}_{T,j}\, v)_i/v_i + \max_i(\mathrm{TAIL}_{T,j}\, v)_i/v_i \leq 0.052 \quad \text{for } T \leq 16 \text{ and } j \leq 32,$$

with monotone but geometrically decaying increments in both T and j, and we extend the bound to a cross-node operator on the SCC obtaining $\rho(M_{\mathrm{cross}}) \leq 0.0366$. The cross-node spectrum is *lower* than the single-node bound at the critical vertex.

We do not claim a proof of the Collatz conjecture. We claim that the structural problem is reduced to two explicit a priori estimates on finite matrices: the asymptotic boundedness of the weighted bound under refinement in T, and the closure of the empirical signature support under refinement in j. We document supporting numerical evidence over a five-decade range of parameters, expose the open estimates as conjectures, and provide all computational scripts as supplementary material.

Contents

*Independent Researcher, Italy. Email: info@pieroborgatta.com. ORCID: 0009-0001-8025-2405.

1 Introduction

The Collatz conjecture asserts that the iteration

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ even,} \\ 3n+1 & \text{if } n \text{ odd,} \end{cases}$$

applied to any positive integer n eventually reaches the trivial cycle $1 \to 4 \to 2 \to 1$. Despite eighty years of attention from combinatorialists, number theorists, and dynamicists, the conjecture remains open. The strongest unconditional progress is due to Tao [1], who established that almost all orbits attain arbitrarily small values relative to the initial point, in a precise density sense.

The bottleneck for a deterministic argument is well known: certain "rebel" starting values produce orbits that climb to many orders of magnitude above their initial value before eventually descending. Empirically these rebels are not isolated curiosities but appear to form structured families which converge in inverse-tree language to common dissipative trunks.

This work proposes that the structural mechanism behind such families is 2-*adic shadowing*: rebel positive integers follow, for finite stretches, the trajectory of negative rational fixed points of the Syracuse map under composition. These "phantoms" have no positive integer realization but locally constrain the ν_2 sequence of nearby positive orbits, producing the observed expansive corridors. Once the constraint is exhausted (necessarily, since positive integers cannot literally equal a negative rational), the orbit drops into a dissipative regime.

We make this framework precise, codify the dynamics as a finite weighted transfer operator on a refined phase quotient, and prove numerically a uniform spectral upper bound which reduces the boundedness of orbits to an explicit a priori estimate on a family of finite matrices.

Structure of the paper. Section 2 fixes notation for the accelerated Syracuse map and the ν_2-debt formalism. Section 3 states and proves the exact congruential shadowing lemma. Section 4 introduces the episode graph and its critical SCC. Section 5 constructs the finite transfer operator at the critical node. Section 6 develops the weighted Collatz–Wielandt perturbative bound. Section 7 reports numerical evidence in T and j. Section 8 extends the bound to a cross-node operator on the entire SCC. Section 9 formulates the explicit conjectures whose resolution would yield a proof. Section 10 is an honest accounting of what this work does and does not establish.

2 Setup: accelerated Syracuse map and gravitational debt

For an odd integer n, the *Syracuse step* is

$$S(n) = \frac{3n+1}{2^{\nu_2(3n+1)}}, \qquad a(n) := \nu_2(3n+1).$$

We write $a_k = a(n_k)$ for the ν_2 sequence along the orbit $n_0, n_1, n_2, \dots$ with $n_{k+1} = S(n_k)$.

A direct expansion gives, after k Syracuse steps,

$$\log_2 n_k = \log_2 n_0 + k \log_2 3 - \sum_{i=0}^{k-1} a_i + O(1).$$

The dominant term $k \log_2 3 - \sum a_i$ is what we call the *gravitational debt* after k steps:

$$D(k) := k \log_2 3 - \sum_{i=0}^{k-1} a_i.$$

Positive $D(k)$ indicates net expansion to step k; the orbit height $\log_2 n_k$ tracks $D(k)$ up to bounded fluctuations.

The threshold $a_i > \log_2 3 \approx 1.585$ separates dissipative from expansive single steps. The Collatz conjecture is morally equivalent to the assertion that $D(k) \to -\infty$ along every positive-integer orbit.

3 Phantom rational orbits and the shadowing lemma

3.1 Phantom words and their rational fixed points

Fix a finite word

$$w = (a_0, a_1, \dots, a_{L-1}) \in \mathbb{N}^L, \qquad A = A(w) := \sum_{i=0}^{L-1} a_i.$$

The composition $S_w := S_{a_{L-1}} \circ \cdots \circ S_{a_0}$ is affine: writing it out one obtains

$$S_w(n) = \frac{3^L n + C_w}{2^A}, \qquad \begin{aligned} C_0 &= 0, \\ C_{j+1} &= 3C_j + 2^{A_j}, \\ A_{j+1} &= A_j + a_j. \end{aligned} \tag{1}$$

The fixed point of S_w in $\mathbb{Q}$ is

$$q_w = \frac{C_w}{2^A - 3^L}. \tag{2}$$

Definition 1 (expansive phantom)**.** The word w is an *expansive phantom* if $3^L > 2^A$. Then $2^A - 3^L < 0$, and assuming $C_w > 0$ (which is automatic from (1)) we have $q_w \in \mathbb{Q}_{<0}$.

The orbit of q_w under S is exactly periodic of length L with ν_2 word w repeated. In the 2-adic completion $\mathbb{Z}_2$, q_w makes sense as a 2-adic integer (since $\gcd(2^A - 3^L, 2) = 1$) and its 2-adic orbit consists of L distinct 2-adic integers cycled by S.

3.2 Exact congruential shadowing

Let w^∞ denote the right-infinite repetition of w, and let $B_m := \sum_{i=0}^{m-1} a_{i \bmod L}$ for $m \geq 0$ be the partial sum of the periodic word.

Lemma 2 (Exact shadowing)**.** *Let w be an expansive phantom with 2-adic representative q_w, and let n be a positive odd integer. If*

$$n \equiv q_w \pmod{2^{B_m+1}} \qquad \textit{(equality in } \mathbb{Z}_2 \textit{ truncated to } B_m + 1 \textit{ bits)},$$

then the first m Syracuse steps of n have ν_2 word exactly $(a_0, a_1, \ldots, a_{m-1})$.

In particular, if $n \equiv q_w \pmod{2^{bA+1}}$, the first b periods (i.e. bL Syracuse steps) of n shadow w^∞.

Proof. We argue by induction on j. Suppose after j Syracuse steps the orbits of n and q_w have followed the same word $(a_0, \ldots, a_{j-1})$. Then both have undergone the same affine transformation $S_{(a_0,\ldots,a_{j-1})}$, which has multiplier $3^j/2^{B_j}$. Therefore

$$S^j(n) - S^j(q_w) = \frac{3^j}{2^{B_j}}(n - q_w).$$

Taking ν_2 of both sides and using $\nu_2(3^j) = 0$,

$$\nu_2(S^j(n) - S^j(q_w)) = \nu_2(n - q_w) - B_j \geq (B_m + 1) - B_j.$$

For step j we need to extract *exactly* a_j factors of 2 from $3S^j(n) + 1$, which agrees with $3S^j(q_w) + 1$ up to valuation

$$\nu_2(3S^j(n) + 1 - 3S^j(q_w) - 1) = \nu_2(3(S^j(n) - S^j(q_w))) \geq B_m + 1 - B_j.$$

For the next valuation to coincide it suffices that

$$B_m + 1 - B_j \geq a_j + 1, \quad \text{i.e.} \quad B_m \geq B_j + a_j = B_{j+1},$$

which holds for all $j < m$ by definition. Induction completes the proof. The "+1" bit is essential: without it one only ensures agreement modulo 2^{a_j}, not the absence of an additional factor of 2. □

Corollary 3 (No infinite shadowing)**.** *A positive integer cannot shadow w^∞ for all m, since that would force $n = q_w$ in $\mathbb{Z}_2$, hence $n = q_w \in \mathbb{Q}_{<0}$, contradicting $n \in \mathbb{N}$.*

4 The episode graph

Lemma 2 shows that an integer n can shadow a phantom w for some maximal number of periods $b = b_w(n)$, after which it must leave the residue class. The natural object for tracking long orbits of integer rebels is therefore not n itself but the sequence of *episodes*

$$n \to (w_1, b_1) \to (w_2, b_2) \to \cdots$$

together with terminal events (drops below the starting value).

4.1 The graph

We monitor a finite set of phantoms identified empirically up to $k \leq 24$ (their explicit list is given in Appendix A). For each pair (phantom k, cycle c, depth b) we associate a *node* (k, c, b). The episode graph G has these nodes as vertices; an edge $(k, c, b) \to (k', c', b')$ records the empirical observation that some integer orbit shadowed (k, c, b) for exactly b periods, then re-entered shadowing of (k', c', b').

4.2 The critical SCC

Empirical exhaustion (script `59_shadow_episode_graph.py`, `60_episode_graph_diagnostics.py`) over the dense sample $b \leq 16$, $t \leq 65535$ identifies a single nontrivial strongly connected component containing

$$(10, 1, b) \text{ for } b \in [1, 15], \quad (11, 1, b) \text{ for } b \in [1, 15], \quad (12, 1, b) \text{ for } b \in [1, 2], \quad (12, 2, b) \text{ for } b \in [1, 2].$$

The *critical node* is $(12, 2, 1)$: it is the unique node of the SCC with significant self-recurrence, and most cross-node transitions terminate at it.

The strategy of the rest of the paper is to prove that this SCC is spectrally subcritical when encoded as a transfer operator with appropriate weights.

5 The critical transfer operator

Fix the critical node $(k, c, b) = (12, 2, 1)$ with phantom word w, period sum A, and 2-adic representative q_w. Every integer n shadowing this node for one period satisfies

$$n \equiv q_w \pmod{2^{A+1}}, \quad \text{i.e.} \quad n = r + t \cdot 2^{A+1}$$

for the unique residue $r = q_w \bmod 2^{A+1}$ and parameter $t \in \mathbb{N}$.

5.1 Refined phase state

For a parameter t, we define the *refined phase state*

$$\sigma(t, h) := (\nu_2(t) \wedge V,\ \operatorname{odd}(t) \bmod 4,\ h \bmod 4),$$

where V is a finite valuation cap, $\operatorname{odd}(t) := t/2^{\nu_2(t)}$, and h is the count of monitored hits accumulated so far. The choice of moduli 4 and 4 is the empirical minimum that makes the spectral truncation stable (script `74_phase_transfer_stress.py`).

5.2 Truncated finite operator

For depths $T \in \mathbb{N}$ and $j \in \mathbb{N}$, we define the truncated transfer operator $\mathrm{FULL}_{T,j}$ on the state space

$$\mathcal{S}_{T,j} := \{\sigma(t, h) : t \in [0, 2^T \cdot j),\ h \in [0, 4)\}$$

as follows. For each source state σ_{src}, enumerate all (t, h) producing it. Trace the Syracuse orbit from $n_0 = r + t \cdot 2^{A+1}$ until either:

1. $n < n_0$ (terminal: contributes no edge);
2. the next monitored hit at the same critical node occurs at $n^* = r + t^* \cdot 2^{A+1}$. This contributes an edge

$$\sigma_{\mathrm{src}} \to \sigma(t^*, h + 1) \quad \text{with weight} \quad 2^{-\delta}, \quad \delta = \lfloor \log_2 t^* \rfloor - \lfloor \log_2 t \rfloor.$$

We aggregate edges and normalize by source counts:

$$(\mathrm{FULL}_{T,j})_{\sigma,\sigma'} = \frac{1}{|\{(t,h) : \sigma(t,h) = \sigma\}|} \sum_{\substack{(t,h) \\ \sigma(t,h)=\sigma}} \mathbf{1}_{\text{trace lands on } \sigma'} \cdot 2^{-\delta(t)}.$$

By construction $\mathrm{FULL}_{T,j}$ is a substochastic-like nonnegative matrix with row sums ≤ 1, the deficit corresponding to terminal transitions.

5.3 Empirical signature decomposition

For each pair (r, h) with $r \in [0, 2^T)$, the j lifts

$$t = r + j' \cdot 2^T, \quad j' \in [0, j),$$

produce j traces. We collect the multiset of *transition signatures* (destination state plus δ bit increment) and declare the *core signature* to be the most frequent one. Edges matching this signature are placed in $\mathrm{CORE}_{T,j}$; all other edges are placed in $\mathrm{TAIL}_{T,j}$. By construction

$$\mathrm{FULL}_{T,j} = \mathrm{CORE}_{T,j} + \mathrm{TAIL}_{T,j}.$$

We refer the reader to script `77_high_bit_tail_bound.py` for the implementation details.

6 Weighted perturbative bound

6.1 Collatz–Wielandt with the Perron eigenvector of FULL

The matrices $\mathrm{FULL}_{T,j}$, $\mathrm{CORE}_{T,j}$, $\mathrm{TAIL}_{T,j}$ are nonnegative. Let $v_{T,j}$ denote a right Perron eigenvector of $\mathrm{FULL}_{T,j}$, normalized so $\|v_{T,j}\|_\infty = 1$. By Perron–Frobenius (after a tiny rank-1 perturbation to ensure irreducibility, see remark below), $v_{T,j}$ exists and is strictly positive.

The decomposition $\mathrm{FULL}_{T,j} = \mathrm{CORE}_{T,j} + \mathrm{TAIL}_{T,j}$ acting on $v_{T,j}$ gives, componentwise,

$$(\mathrm{FULL}_{T,j} v_{T,j})_i = (\mathrm{CORE}_{T,j} v_{T,j})_i + (\mathrm{TAIL}_{T,j} v_{T,j})_i.$$

Taking the maximum over i such that $(v_{T,j})_i > 0$ and using $\rho(\mathrm{FULL}_{T,j}) = \max_i (\mathrm{FULL}_{T,j} v_{T,j})_i / (v_{T,j})_i$,

$$\boxed{\rho(\mathrm{FULL}_{T,j}) \;\le\; \underbrace{\max_i \frac{(\mathrm{CORE}_{T,j} v_{T,j})_i}{(v_{T,j})_i}}_{\text{core action}} \;+\; \underbrace{\max_i \frac{(\mathrm{TAIL}_{T,j} v_{T,j})_i}{(v_{T,j})_i}}_{\text{tail action}}\;.} \tag{3}$$

We denote the right-hand side $\mathrm{bound}(T, j)$.

Remark 4 (numerical implementation)**.** We compute $v_{T,j}$ by power iteration on $\mathrm{FULL}_{T,j} + \varepsilon J$ with $\varepsilon = 10^{-12}$ and J the all-ones matrix. The rank-1 perturbation removes reducibility numerically without affecting the dominant eigenvalue beyond ε. Convergence is checked in ℓ^∞ norm to tolerance 10^{-12}.

Remark 5 (why CORE's Perron is wrong)**.** Using the Perron eigenvector of $\mathrm{CORE}_{T,j}$ instead of $\mathrm{FULL}_{T,j}$ yields a divergent "bound": $\mathrm{CORE}_{T,j}$ is typically reducible, with its Perron eigenvector concentrated on a strict subset of states, and $\mathrm{TAIL}_{T,j}$ routinely sends mass to states outside that subset. The Collatz–Wielandt ratios then blow up. The empirical fix and the theoretical fix both point to using $v_{T,j}$ from $\mathrm{FULL}_{T,j}$.

7 Numerical evidence

7.1 Bound stability in T

We compute bound(T, j) for $T \in \{8, 10, 12, 14, 16\}$ and $j \in \{16, 32\}$ using script `84_weighted_perturbation_b`
(with the eigenvector selection from the previous remark). Results:

T	j	ρ(FULL)	core action	tail action	bound
8	16	0.0294	0.0114	0.0294	0.0408
8	32	0.0299	0.0102	0.0299	0.0400
10	16	0.0303	0.0184	0.0303	0.0487
10	32	0.0308	0.0177	0.0308	0.0485
12	16	0.0303	0.0137	0.0303	0.0499
12	32	0.0306	0.0135	0.0306	0.0497
14	16	0.0306	0.0157	0.0306	0.0511
14	32	0.0307	0.0154	0.0307	0.0507
16	16	0.0307	0.0169	0.0307	0.0516

The bound is monotonically increasing in T but the increments decay geometrically:

$$\Delta_8^{10} = +0.0079, \quad \Delta_{10}^{12} = +0.0012, \quad \Delta_{12}^{14} = +0.0012, \quad \Delta_{14}^{16} = +0.0005.$$

A geometric extrapolation yields

$$\lim_{T\to\infty} \text{bound}(T, 16) \leq 0.054.$$

The bound is essentially independent of j: differences between $j = 16$ and $j = 32$ at fixed T never exceed 0.001.

7.2 Spectral gap ρ(FULL) $-$ ρ(CORE)

Independently, the spectral radii of FULL and CORE are computed (script `83_perturbation_gap_scaling.py`
The gap ρ(FULL) $-$ ρ(CORE) shrinks with T:

$$T = 6: \ 0.025\text{–}0.032, \quad T = 8: \ 0.025\text{–}0.028, \quad T = 10: \ 0.018\text{–}0.019,$$

$$T = 12: \ 0.016\text{–}0.017, \quad T = 14: \ 0.015, \quad T = 16: \ 0.014.$$

The spectral mass of FULL migrates progressively to CORE as the quotient is refined.

7.3 Truncation stability in j (Cauchy property)

For fixed T, we check whether FULL$_{T,j}$ admits a Cauchy property in j via two independent metrics (script `85_truncation_stability.py`):

- **signature drift**: the fraction of (r, h) groups whose dominant signature changes between consecutive doublings of j;
- **bound increment**: bound$(T, 2j)$ $-$ bound(T, j).

Results:

T	j	sig_drift_vs_prev	bound	Δ_{prev}
8	16	–	0.0408	–
8	32	0.0078	0.0400	−0.0007
8	64	0.0078	0.0421	+0.0021
8	128	0.0000	0.0425	+0.0004
10	16	–	0.0487	–
10	32	0.0039	0.0485	−0.0003
10	64	0.0088	0.0509	+0.0024
10	128	0.0000	0.0515	+0.0006

Two facts:

1. The signature drift drops to *exactly zero* when j is doubled from 64 to 128, for both $T \in \{8, 10\}$. The empirical transition signatures of all (r, h) groups stabilize.

2. The bound increments decay by a factor ~ 4 per doubling, suggesting a Cauchy bound $\lim_{j\to\infty} \text{bound}(T = 10, j) \lesssim 0.052$.

Combining the two extrapolations, we conjecture

$$\sup_{T,j} \text{bound}(T, j) < 0.06,$$

a margin of $16\times$ from the criticality threshold 1.

8 Cross-node certificate on the SCC

The single-node bound at the critical node is the worst case: running script `86_scc_node_certificates.py` at $T = 10, j = 32$ across the SCC nodes yields:

node	ρ(FULL)	bound	status
$(10, 1, 1)$	≈ 0	0.0000	OK (return-to-self vacuous)
$(10, 1, 2)$	≈ 0	0.0000	OK
$(11, 1, 1)$	≈ 0	0.0000	OK
$(11, 1, 2)$	≈ 0	0.0000	OK
$(12, 1, 1)$	0.0035	0.0043	OK
$(12, 2, 1)$	0.0308	0.0485	OK (worst case)
$(12, 2, 2)$	0.0007	0.0007	OK

The nodes $(10, 1, *)$, $(11, 1, *)$, $(12, 2, 2)$ have nearly vanishing return-to-self spectrum. They are entry gates to the SCC, not self-recurrent.

8.1 The cross-node operator

We construct the assembled cross-node operator M_{cross} on the union state space

$$\bigsqcup_{(k,c,b)\in\text{SCC}} \mathcal{S}_{T,j}^{(k,c,b)}.$$

Edges are recorded between the source state at one node and the *first hit* state at any node of the SCC, with weight $2^{-\delta_{\text{bit}}}$ where δ_{bit} is the bit-length increment between source n_0 and destination value (a node-invariant quantity). The construction is implemented in script `87_cross_node_transfer.py`.

At $T = 8, j = 8$ on 5 representative SCC nodes:

$$\rho(M_{\text{cross}}) = 0.0366, \qquad \text{bound}(M_{\text{cross}}) = 0.0366.$$

The cross-coupling *lowers* the spectrum from the single-node worst case (0.0485). Topologically the SCC is a star centered on $(12, 2, 1)$:

```
(10,1,1) -> (12,2,1)   8128  (gate)
(11,1,1) -> (12,2,1)   8192  (gate)
(12,2,2) -> (12,2,1)   8128  (gate)
(12,2,1) -> (12,2,1)    812  (only significant self-loop)
(12,2,1) -> (12,1,1)    308
(12,1,1) -> (12,1,1)     96
(12,2,2) -> (12,2,2)     64
(12,2,1) -> (12,2,2)      8
```

The single-node bound was therefore overly conservative; including cross-node transitions strictly improves it.

9 Conjectures and the spectral reduction statement

We isolate the open analytic content as two explicit conjectures.

Conjecture 6 (uniform bound in T)**.** There exists a constant $B_\infty < 1$ such that

$$\sup_{T\in\mathbb{N},\ j\in\mathbb{N}} \text{bound}(T, j) \le B_\infty.$$

Empirically the data support $B_\infty \approx 0.06$.

Conjecture 7 (signature closure in j)**.** For each (T, r, h) there exists $j^*(T, r, h)$ such that the dominant transition signature stabilizes for all $j \ge j^*$, and furthermore

$$\sup_{j\ge j^*} \left|\text{bound}(T, j) - \text{bound}(T, j^*)\right| \le C \cdot 2^{-\beta(j-j^*)}$$

for some $C, \beta > 0$ uniform in (T, r, h).

Theorem 8 (conditional reduction, sketch)**.** *Assume Conjectures 6 and 7. Then for every positive integer n, the Syracuse orbit $S^k(n)$ satisfies $\liminf_{k\to\infty} S^k(n) = 1$.*

The argument is the standard one for transfer-operator subcriticality combined with the no-infinite-shadowing Corollary 3: assuming the conjectures, the critical SCC has spectral radius $\le B_\infty < 1$ on a refined quotient that captures all dynamically relevant phases, hence the return-to-critical mass strictly contracts, hence orbits cannot remain forever near the SCC, hence by exhaustion of the empirical phantom set every orbit eventually escapes to a dissipative regime with average $\nu_2 > \log_2 3$ and descends. The full argument requires an exhaustive enumeration of phantoms below a controlled threshold and a verification that no further SCCs appear; see Section 10 for an honest discussion of the gaps.

10 What this is and what this is not

This work is a *program*, not a proof. We summarize what is established, what is empirical, and what is missing.

Established (rigorously).

- Lemma 2 (exact congruential shadowing) is a theorem with elementary proof. It identifies the precise residue condition for b-fold phantom shadowing.
- Corollary 3 (no infinite shadowing) follows formally.
- The construction of the finite transfer operators $\mathrm{FULL}_{T,j}$, $\mathrm{CORE}_{T,j}$, $\mathrm{TAIL}_{T,j}$ at the critical node is a deterministic algorithm; the matrices are exactly computable for all finite T, j.
- The bound (3) is a Collatz–Wielandt consequence of nonnegativity; it gives a true upper bound on $\rho(\mathrm{FULL}_{T,j})$ for each finite (T, j).

Empirical (numerically supported, not proved).

- All numerical bounds reported in Sections 7 and 8.
- The geometric decay of bound increments in T and j.
- The closure of dominant signatures at $j = 128$ for $T \in \{8, 10\}$.
- The identification of $(12, 2, 1)$ as the worst-case node of the SCC.
- The structural "star" topology of the SCC around $(12, 2, 1)$.

Missing.

- Proofs of Conjectures 6 and 7. We believe the natural framework is transfer-operator analysis on a suitable Banach space of 2-adic Lipschitz functions; a Lasota–Yorke inequality for the family $\{\mathrm{FULL}_T\}$, combined with Hennion's theorem and Keller–Liverani perturbation theory [4], would yield Conjecture 6. We have not carried this out.
- Verification that the empirical phantom set ($k \leq 24$) is complete in the sense that no SCC outside the listed nodes contributes. Empirical sweeps to $b \leq 16$, $t \leq 65535$ find none, but this is not a proof.
- A formal version of Theorem 8.

Honest claim. We have reduced the open structural content of the Collatz conjecture to a problem of *spectral perturbation theory on a finite explicit family of nonnegative matrices*, supported by numerical evidence with a $16\times$ safety margin. We do not claim the conjecture itself.

Methodology and AI disclosure

This work was carried out by the author in iterative collaboration with multiple large language model assistants. A clear separation between contributions is essential for the reader's correct evaluation of the material.

Author's contribution. The conceptual framing is due to the author: the gravitational debt metaphor, the empirical identification of $\nu_2 = 1$ corridors as the locus of expansive behavior, the structural intuition that integer rebels shadow negative rational fixed points of S, the strategic pivot from the empirical Lyapunov candidates of the earlier phase to the present spectral program, and the decision to publish at this stage. The author has no formal mathematical training and made no direct technical contribution to the formalization, the implementation, or the proofs.

AI assistance. The Python scripts in `scripts/early_empirical/` (the quasi-Lyapunov phase) were developed with assistance from OpenAI Codex and Google Gemini. The Python scripts in `scripts/spectral_program/` (the present spectral reduction), the mathematical formalization of Lemma 2 and the weighted Collatz–Wielandt bound of Section 6, and the LaTeX drafting of this paper were carried out with substantial assistance from Anthropic's Claude (via the Claude Code interface).

Critical limitation. As of the present version, no human mathematical expert has reviewed the proofs, the formal definitions, or the construction of the transfer operator. The numerical results are deterministic and reproducible from the supplementary scripts and may be checked independently. The mathematical claims, however, rest on AI-generated formalization that has not been externally validated. Subtle errors in definitions, inductive arguments, or quantifier manipulations cannot be excluded. Readers are invited, and indeed urged, to scrutinize Sections 3, 5, and 6 with particular care, and the author explicitly welcomes correspondence from researchers in symbolic dynamics, p-adic dynamical systems, and transfer operator theory before any portion of this material is treated as established.

References

[1] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, arXiv:1909.03562, 2019; published in Forum of Mathematics, Pi, 2022.

[2] J. C. Lagarias, *The 3x+1 problem and its generalizations*, American Mathematical Monthly 92 (1985), 3–23.

[3] J. C. Lagarias (ed.), *The Ultimate Challenge: The 3x+1 Problem*, American Mathematical Society, 2010.

[4] G. Keller and C. Liverani, *Stability of the spectrum for transfer operators*, Annali della Scuola Normale Superiore di Pisa, Cl. Sci. (4) 28 (1999), 141–152.

[5] V. Baladi, *Positive Transfer Operators and Decay of Correlations*, World Scientific, 2000.

[6] V. Anashin, *The non-Archimedean theory of discrete systems*, Math. Comput. Sci. 6 (2012), 375–393.

A Phantom records used in this work

The expansive phantom representatives identified empirically up to $k = 24$ and used as monitored nodes in Section 4:

k	$q_w \approx$
10	$-5\,739\,192\,681\,529/2\,404\,426\,874\,857 \approx -2.387$
11	$-2\,199\,012\,694\,031/709\,849\,655\,971 \approx -3.098$
12 (cycle 1)	$-2123/1675 \approx -1.267$
12 (cycle 2)	$-817/601 \approx -1.359$
20	$-104\,312\,644\,103/30\,307\,317\,785 \approx -3.442$

The full list and the construction of these representatives from the rational fixed point formula (2) is implemented in `54_phantom_rational_shadows.py`.

B Supplementary scripts

All scripts referenced in this paper are available as supplementary material in the GitHub repository

https://github.com/PieroBorgatta/Collatz

and as a frozen Zenodo deposit

https://doi.org/10.5281/zenodo.20021538

The principal scripts and what each computes:

`54_phantom_rational_shadows.py`
Computes the 2-adic representatives of expansive phantom cycles; underpins all subsequent constructions.

`55_shadowing_congruence.py`
Verifies Lemma 2 on the empirical phantom set for $b \leq 10$.

`59_shadow_episode_graph.py`, `60_episode_graph_diagnostics.py`
Constructs the episode graph and identifies its SCCs.

`75_critical_symbolic_operator.py`
Builds the truncated finite operator at the critical node.

`77_high_bit_tail_bound.py`
Decomposes FULL = CORE + TAIL from the empirical signature majority.

`82_boundary_scc_spectral_scaling.py`
Spectral scaling on SCCs touching the boundary layer $\nu_2(t) \geq T$.

`83_perturbation_gap_scaling.py`
Measures the gap $\rho(\text{FULL}) - \rho(\text{CORE})$.

`84_weighted_perturbation_bound.py`
Computes the weighted bound (3).

`85_truncation_stability.py`
Cauchy/signature stability of the truncation in j.

`86_scc_node_certificates.py`
Per-node bounds on the SCC.

`87_cross_node_transfer.py`
Assembled cross-node transfer operator on the SCC.